MARKING SCHEME

Model Paper — 03

Third Term Test (Full Syllabus)

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Note (for teachers):

- Part A: 2 marks for each question (generally 1 mark for the correct method and 1 mark for the correct final answer). — Total 40 marks
- Part B: 10 marks for each question. — Total 60 marks. The marks for each step are shown in the right-hand column.
- Award full marks for correct alternative methods. Accept a reasonable range for values read from the graph.

Part A — Answers

1. $\sqrt{7744}$: $77 \rightarrow 8^2 = 64$, remainder 13 ; $1344 \rightarrow 168 \times 8 = 1344$, remainder 0

Final answer: 88

(2 marks)

2. $\frac{7}{4} + \frac{10}{4} = \frac{17}{4}$; $\frac{17}{4} \div \frac{17}{10} = \frac{17}{4} \times \frac{10}{17} = \frac{10}{4}$

Final answer: $2\frac{1}{2}$

(2 marks)

3. $(x^2 + 6x + 9) - (x^2 - 6x + 9)$

Final answer: $12x$

(2 marks)

4. $3x^2 + 12x - 2x - 8 = 3x(x + 4) - 2(x + 4)$

Final answer: $(3x - 2)(x + 4)$

(2 marks)

5. $x^2 - 4 = (x - 2)(x + 2)$; $x^2 + x - 6 = (x + 3)(x - 2)$

Final answer: $(x - 2)(x + 2)(x + 3)$

(2 marks)

6. $\frac{2x}{(x - 1)(x + 1)} \times \frac{x + 1}{4x^2}$

Final answer: $\frac{1}{2x(x - 1)}$

(2 marks)

7. $\frac{60}{360} \times 2 \times \frac{22}{7} \times 10.5 = \frac{1}{6} \times 66$

Final answer: 11 cm

(2 marks)

8. 9 parts = 180° ⇒ one part = 20° ; 4 × 20°

Final answer: 80°

(2 marks)

9. $x + 1 = 3^2 = 9$

Final answer: x = 8

(2 marks)

10. $\lg 3^2 + \lg 5 - \lg 9 = \lg 9 + \lg 5 - \lg 9$

Final answer: lg 5

(2 marks)

11. $S_{20} = \frac{20}{2} [2 \times 2 + 19 \times 3] = 10 \times 61$

Final answer: 610

(2 marks)

12. $n(A \cup B) = 30 + 25 - 12 = 43$; $50 - 43$

Final answer: 7

(2 marks)

13. $100 \rightarrow 120 \rightarrow 120 \times 0.8 = 96$

Final answer: a decrease of 4%

(2 marks)

14. Distance = 60 × 4 = 240 km ; 240 ÷ 3

Final answer: 80 km/h

(2 marks)

15. $\frac{T}{2\pi} = \sqrt{\frac{l}{g}} \Rightarrow \frac{T^2}{4\pi^2} = \frac{l}{g}$

Final answer: $l = \frac{gT^2}{4\pi^2}$

(2 marks)

16. $-2x \leq 6 \Rightarrow x \geq -3$

Final answer: −3

(2 marks)

17. Multiples of 4: 4, 8, 12, 16, 20 (5) ; $\frac{5}{20}$

Final answer: $\frac{1}{4}$

(2 marks)

18. Sum = $5 \times 12 = 60$; $60 - 20 = 40$; $40 \div 4$

Final answer: 10

(2 marks)

19. $\Delta DBC = \Delta ABC = 35\ cm^2$ (same base, between the same parallels) ; DE is a median $\Rightarrow \Delta DEC = \frac{1}{2} \times 35$

Final answer: 17.5 cm^2

(2 marks)

20. $0.9\ m^3 = 900$ litres ; $900 \div 12 = 75$ minutes

Final answer: 1 hour 15 minutes

(2 marks)

Part B — Answers and Mark Allocation

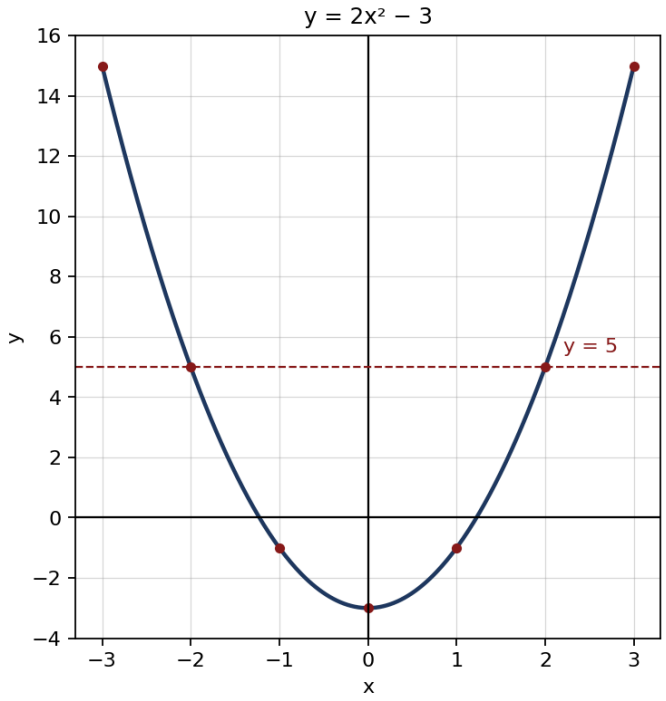
Question 1

(a)	
$\frac{(x-3)(x+2)}{(x-3)(x+3)} \times \frac{x(x+3)}{x(x+2)}$	**2**
$= 1$	**1**
(b)	
$x + y = 47$ .....①	**1**
$2x - 3y = 9$ .....②	**1**
① × 2 − ②: $5y = 85 \Rightarrow y = 17$	**1**
$x = 30$; the numbers are 30 and 17	**1**
(c)	
$(3x - 1)(x - 2) = 0$	**2**
$x = \frac{1}{3}$ or $x = 2$	**1**

Total marks: 10

Question 2

(a)	
$x = -2 \Rightarrow y = 5$; $x = 1 \Rightarrow y = -1$	**1**
(b)	
Correct scale and axes	**1**
Plotting the 7 points correctly	**1**
Smooth curve	**1**



(c)	
(i) (0, −3)	**1**
(ii) Reading the points where the curve cuts the x-axis	**1**
$x \approx -1.2$ or $x \approx 1.2$ (accept ±1.2 to ±1.3)	**1**
(iii) from the line $y = 5$: $-2 < x < 2$	**1**
(iv) $x < 0$	**1**
(d)	
$y = 2x^2$	**1**

Total marks: 10

Question 3

(a)	
Cross-sectional area = $\frac{1}{2} \times 7 \times 24 = 84\ \text{cm}^2$	**1**
Volume = $84 \times 11 = 924\ \text{cm}^3$	**1**
(b)	
Hypotenuse = $\sqrt{(7^2 + 24^2)} = \sqrt{625} = 25\ \text{cm}$	**1**
Rectangular faces = $(7 + 24 + 25) \times 11 = 616\ \text{cm}^2$	**1**
Total surface area = $616 + 2 \times 84 = 784\ \text{cm}^2$	**1**
(c)	
$\frac{22}{7} \times 7 \times 7 \times h = 924$	**1**
$154h = 924$	**1**
$h = 6\ \text{cm}$	**1**
(d)	
$2\pi rh = 2 \times \frac{22}{7} \times 7 \times 6$	**1**
$= 264\ \text{cm}^2$	**1**

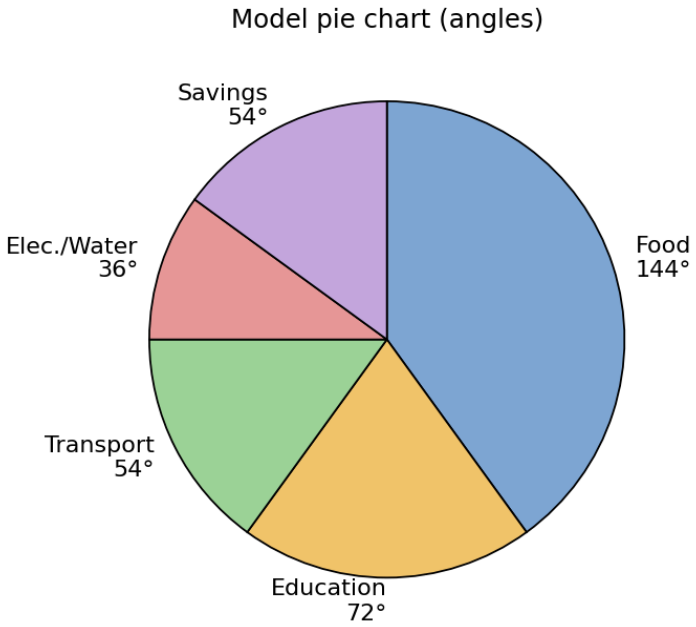
Total marks: 10

Question 4

(a)	
Correct rough sketch	**1**
(b)	
ΔABE and $ABCD$ stand on the same base AB and lie between the parallel lines $AB \parallel DC$	**1**
$\therefore \Delta ABE = \frac{1}{2} \times ABCD$	**1**
(c)	
$\Delta ADE + \Delta BCE = ABCD - \Delta ABE = ABCD - \frac{1}{2} ABCD = \frac{1}{2} ABCD$	**1**
$= \Delta ABE$	**1**
(d)	
$\Delta ADE + \Delta BCE = \frac{1}{2} \times 84 = 42\ cm^2$	**1**
The two triangles have equal heights (the distance between AB and DC) $\Rightarrow$ ratio of areas = DE : EC = 3 : 4	**1**
$\Delta ADE = \frac{3}{7} \times 42 = 18\ cm^2$; $\Delta BCE = \frac{4}{7} \times 42 = 24\ cm^2$	**1**
(e)	
DG is a median of triangle ADE $\Rightarrow \Delta DGE = \frac{1}{2} \Delta ADE$	**1**
$= 9\ cm^2$	**1**

Total marks: 10

Question 5

(a)	
Savings = 60 000 − 51 000 = Rs. 9 000	**1**
Food 144°, Education 72°, Transport 54°	**1**
Electricity and water 36°, Savings 54°	**1**
(b)	
	
Correct radius and angles (± 2°)	**2**
Labelling the sectors	**1**
(c)	
$\frac{24\ 000}{60\ 000} \times 100 = 40\%$	**1**
(d)	
Increase = $24\ 000 \times \frac{15}{100}$ = Rs. 3 600	**1**
New savings = 9 000 − 3 600 = Rs. 5 400	**1**
Angle = $\frac{5\ 400}{60\ 000} \times 360° = 32.4°$	**1**

Total marks: 10

Question 6

(a)	
Man-days in 12 days = 20 × 12 = 240	**1**
240 man-days ⇒ ⅓ of the wall ; whole wall = 240 × 3	**1**
= 720 man-days	**1**
(b)	
Remaining work = 720 − 240 = 480 ; 480 ÷ 20 = 24 days	**1**
Days remaining = 18 ⇒ 24 − 18 = delayed by 6 days	**1**
(c)	
Workers required = 480 ÷ 18 = 26.67	**1**
⇒ at least 27 workers	**1**
Newly employed = 27 − 20 = 7 workers	**1**
(d)	
480 ÷ 25 = 19.2 ⇒ takes 20 days	**1**
20 − 18 = delayed by 2 days	**1**

Total marks: 10